

VERIQO Real-World Network Model (P3)

The four real-world networks VERIQO must eventually connect to — maritime, insurance, commodity, and financial — with, for each, what exists in code today and what genuinely does not.

Nothing described as a live integration exists. Every connector in this repository is a contract plus a *simulated* source. That is stated in each package's own doc comment, and — more usefully — enforced in code: the simulated connectors emit only `ModeSynthetic` and **have no code path capable of producing `ModeLive`**. A synthetic feed cannot be mistaken for a real one by accident.

1. What blocks these integrations

Not engineering. Each requires something only a commercial relationship produces:

- A **data contract** with a provider (AIS aggregator, market data vendor, customs broker, carrier).
- A **counterparty agreement** — an insurer, a bank, a terminal operator willing to exchange evidence.
- **Platform membership** — eBL platforms operate under rulebooks with admission requirements.
- **Regulatory standing** for some sources (customs, payment rails).

This is why P3 is documented rather than built. Writing a connector against an API you have no contract to call produces untested code shaped by a guess at the payload.

2. The nine questions every integration must answer

Applied uniformly. An integration that cannot answer all nine is not ready to carry evidence, however well it moves data.

#	Question	Why it decides evidential value
1	Source — who is the authoritative origin?	An aggregator is not the origin. Provenance must record the actual origin, not the last hop.
2	Contract — what are the licensing, redistribution, and retention terms?	Evidence you may not retain or show a counterparty is not evidence you can use in a dispute.
3	Identity — how is the source authenticated, and how are entities it names resolved?	A vessel ID from an unauthenticated feed is an assertion, not a fact.
4	Acquisition — push, pull, batch, streaming? What are the gaps?	Gap behaviour determines whether absence means "did not happen" or "we were not listening".
5	Integrity — what is hashed, and at what point?	Hash at acquisition, before any normalization, or the hash attests to your transformation rather than their data.
6	Provenance — what is recorded about how it arrived?	Must survive into the custody chain and be legible to a challenger years later.
7	Retention — how long, under whose obligation, and what happens on legal hold?	Provider terms and legal-hold obligations conflict more often than expected.
8	Reconciliation — how is it checked against another source?	Single-source evidence is assertion. Reconciliation is where contradiction gets detected.
9	Failure handling — what happens on outage, malformed data, or provider dispute?	Silent gaps are the worst failure mode: they look like normal quiet.

The existing connector contracts already implement the machinery for questions 4–6 and 9. Questions 1–3, 7, and 8 are where a real integration does its distinctive work.

3. Maritime

Integration	Built	Not built
AIS	A full streaming connector with reconnect/backoff, a wire-message schema, and hand-written synthetic fixtures explicitly not captured from any real feed. No test dials a real address.	Any commercial AIS provider contract.
Port / terminal events	Nothing specific.	Terminal operating system integrations; each major terminal differs.
Vessel identity / registry	Vessel identity is recorded as an asserted string.	Resolution against IMO/flag-state registries. Today an identifier proves it was asserted, not that it is correct.
eBL platforms	A Bill of Lading / customs ingestion contract with structural and semantic validation (instrument type known, container numbers well-formed).	Membership of any eBL platform, and its rulebook obligations.

Distinctive hard problem: AIS is *reported by the vessel*, and can be switched off, spoofed, or misconfigured. Its evidential value is therefore in **corroboration and contradiction** — an AIS gap alongside a port record is a finding — not in treating position reports as ground truth. Any maritime integration that presents AIS as fact rather than as one attested source has misunderstood the product.

4. Insurance

Integration	Built	Not built
Claims / policy evidence	An ingestion contract with validation and a simulated connector, honestly labeled.	Any insurer or claims-management-system connection.
Broker / adjuster / surveyor	The evidence model carries <code>party_id</code> and <code>evidence_kind</code> (SURVEY, ADJUSTER_REPORT, FNOL, INVOICE).	Party identity verification; submission workflows for third-party professionals.
P&I clubs	Nothing specific.	Club-specific reporting requirements.
Claims data	Full case → decision → action → dossier pipeline.	Bordereaux formats, claims-system schemas.

Distinctive hard problem: the parties are **structurally adversarial at the margin** — insurer, insured, broker, and surveyor have partly opposed interests. This is where VERIQO's independent verifiability is worth most: each party can verify the record without trusting the others. It is also where the trusted-key-registry question becomes operationally unavoidable rather than optional, because "who signed this" is precisely what is in dispute.

5. Commodity

Integration	Built	Not built
B/L and customs	The BoL contract: parsing, structural and semantic validation.	Any carrier or customs EDI feed; this sandbox has no reachable path to one.
Trader systems	Nothing specific.	CTRM integrations.

Integration	Built	Not built
Refinery / terminal	Nothing specific.	Quantity and quality certificate ingestion.
Cargo data	Trade domain metadata (<code>document_type</code> , <code>transfer_event_id</code> , <code>holder_identity</code>).	Assay results, inspection certificates, loss reports.

Distinctive hard problem: commodity evidence is heavily **measurement-based** — quantity and quality figures produced by instruments and inspectors, where disputes turn on *whose measurement* and *taken how*. The provenance model must capture instrument identity and method, not merely a number and a source name. The evidence model supports this; no integration exercises it yet.

6. Financial

Integration	Built	Not built
Payment evidence	A payment/transaction ingestion contract with a simulated connector; explicitly no path to SWIFT, ACH, or any card network.	Any real payment rail.
Trade finance / banks	Nothing specific.	LC and guarantee lifecycle integration.
Collateral / transaction evidence	The generic evidence pipeline applies.	Collateral management systems.

Distinctive hard problem: the **regulatory perimeter**. Payment and trade-finance data carries AML, sanctions, and banking-secrecy obligations that constrain what may be retained and shown. VERIQO performs **no** sanctions or watch-list screening and asserts no such capability. An integration here needs compliance design before engineering design, and the retention/legal-hold model must be reconciled with obligations that may *require* deletion and *prohibit* it in the same dataset.

7. What a first real integration should look like

Not the largest network — the one where all nine questions are answerable:

1. **One source, one counterparty.** A single provider with a real contract, feeding a single pilot customer's real workflow.
2. **Answer the nine questions in writing before writing code.** The answers determine the connector's shape; discovering them afterward means rewriting it.
3. **Hash at acquisition**, before normalization, so integrity attests to what the provider sent rather than to your processing of it.
4. **Reconcile against something.** A single unreconciled source produces attested assertions. The second source is where contradictions — the thing customers actually pay to detect — become visible.
5. **Make gaps explicit.** Record feed outages as events. Silence that is not recorded as silence will later be read as evidence that nothing happened.
6. **Keep synthetic and live structurally distinguishable**, exactly as the existing connectors do. A live-mode flag that a synthetic path can never set is worth more than a convention people are asked to respect.

8. Honest summary

The **contract layer** for external ingestion is real, tested, and reusable: validation, state machines, failure handling, and a synthetic/live separation enforced by the type system rather than by discipline.

The **live provider integrations are entirely absent**, in all four networks, and are blocked on commercial relationships rather than engineering capacity.

Any material describing VERIQO's real-world connectivity must say this. Describing the connector contracts as "AIS integration" or "payment integration" without the word *simulated* would be exactly the kind of overstatement the connectors' own doc comments were written to prevent.